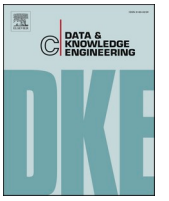
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An explainable machine learning approach for automated medical decision support of heart disease

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ABSTRACT

Coronary Heart Disease (CHD) is the dominant cause of mortality around the world. Every year, it causes about 3.9 million deaths in Europe and 1.8 million in the European Union (EU). It is responsible for 45 % and 37 % of all deaths in Europe and the European Union, respectively. Using machine learning (ML) to predict heart diseases is one of the most promising research topics, as it can improve healthcare and consequently increase the longevity of people's lives. However, although the ability to interpret the results of the predictive model is essential, most of the related studies do not propose explainable methods. To address this problem, this paper presents a classification method that not only exhibits reliable performance but is also interpretable, ensuring transparency in its decision-making process. SHapley Additive exPlanations, known as the SHAP method was chosen for model interpretability. This approach presents a comparison between different classifiers and parameter tuning techniques, providing all the details necessary to replicate the experiment and help future researchers working in the field. The proposed model achieves similar performance to those proposed in the literature, and its predictions are fully interpretable.

1. Introduction

The heart is responsible for circulating blood through the network of arteries and veins. It receives oxygenated blood from the coronary arteries, which then pump deoxygenated blood to the lungs. In the lungs, the blood charges oxygen and discharges carbon dioxide, a waste product of metabolism [1]. Cardiovascular disease (CVD) is a term used to describe problems affecting the heart or blood vessels. Every year, it is responsible for causing about 3.9 and 1.8 million deaths in Europe and the European Union (EU), respectively. Consequently, CVD is accountable for 45 % and 37 % of all deaths causes in Europe and the EU. [2]. In Europe, CVD has a great impact on economies with an annual cost of 196 billion euros. This corresponds to about 54 % and 24 % caused by direct health care and lost productivity, respectively [3]. Coronary heart disease (CHD) happens when the blood supply is affected in the coronary arteries. This process is called atherosclerosis and is a life-threatening condition where fat substances known as plaques, or atheroma, obstruct arteries. It is the major cause of mortality worldwide causing approximately one-third of deaths worldwide [4] and the most common is cardiovascular illness [5]. It is expected that in the coming years, the incidence of the disease will continue to increase, mainly due to the general aging of the population, but also due to the increase in obesity and diabetes [6].

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Related factors such as smoking, obesity, high blood pressure, lack of exercise regularly, and high cholesterol are some of the risks that can lead to the development of atherosclerosis. The most frequent symptoms are chest pain or discomfort (angina) and difficulties in breathing [7]. Finally, it can lead to heart attacks or heart failure if there is no early detection of the problem.

Traditional medical diagnosis is obtained through the analysis of the patient's history and medical tests [8]. It is a public health problem, people can be diagnosed as medical patients due to mistakes in data analysis, leading to a possible treatment that will not bring any benefit [9]. Overdiagnosis is a "wrong" diagnosis of a disease that will never cause the patient to manifest any kind of symptoms or cause their death. Some heart failure patients may suffer from the problem of overdiagnosis [10]. Furthermore, machine learning (ML) can eliminate some problems caused by human factors such as lack of concentration, fatigue, costly, time-consuming processes, and even association or decision errors [11]. ML algorithms in health systems can lead to human cognitive enhancement and consequently to better results [12]. The number of disease prediction systems supported by ML has been increasing and proving their effectiveness as a useful tool to support medical decision-making [13]. Different approaches have been made in recent years. Ensemble learning classification has been tested in four different CHD datasets [14]. In Machine Intelligence Framework for Heart (MIFH) Disease Diagnosis system [15], the authors used factor analysis of mixed data (FAMD) techniques for feature extraction and then applied different tuned ML models.

Every ML-based system must provide explanations for their outcome in a manner that can be understood by non-technical domain experts [16]. When applying ML in medicine it is imperative to focus on the model performance but also provide interpretability. Decision-makers must understand how the classifier behaves at a given result to trust that prediction [17,18]. The need to understand the model behavior has given rise to an area of research called explainable AI (XAI). One of the most widely used XAI techniques is the SHapley Additive exPlanations (SHAP) method. It consists of a game theoretic approach to explain the output of any ML model, examining how the model's output is connected to each input variable [19]. Several works have explored its application on healthcare related domains [16,19,20]. Therefore, the authors proposed an ML approach for predicting CHD using the IEEEDataPort CHD database [21]. This study aims to automate, improve, and optimize the detection of CHD diseases to increase the patient's chances of survival. It also emphasizes the need for proper data modeling techniques that ensure both high performance and interpretability of predictions. Additionally, achieving a shared understanding of the domain under investigation is also essential, as it supports the design of effective information systems, which is crucial for advancing data-driven decision support systems in healthcare [22]. The main contribution of this work is the creation of several ML classifiers exploring different tuning techniques and using the SHAP method to interpret the predictions. On the one hand, it is difficult to find interpretable approaches to this problem in current literature studies. On the other hand, the proposed methods available in the state-of-the-art focus on enhanced performance and fail to explain each prediction made. The proposed method includes a classification model that not only has reliable performance but is also fully interpretable.

This document is organized as follows: Section 2 describes the dataset, data pre-processing steps, ML configuration, and statistical analysis. In Section 3, we present all the results obtained and the explanation of the model predictions. Section 4 shows the discussion made where we analyze the results obtained, compare them with previous studies done in the area, and show our contributions. Finally, Section 5 presents the conclusion of the work.

2. Methodology

2.1. Dataset population and statistical analysis

The dataset was obtained from the IEEEDataPort database [21]. It is a combination of five different heart disease datasets: Cleveland, Hungarian, Switzerland, Long Beach VA, and Statlog (Heart). These datasets are the most commonly used in the literature to create predictive models using ML [23]. The different datasets were combined using eleven common features and the resulting dataset has 1190 instances in total. This data has been collected and combined into a single set to enable research that will lead to advances in clinical diagnosis and the early detection and treatment of patients. It has in total, 23.6 % (281) female patients and 76.4 % (909) male patients. The target class is almost balanced with 47.1 % (561) patients without heart diseases and 52.9 % (629) patients with heart diseases. The imbalance ratio is 0.89. Detailed information on the data is presented in Appendix A. The population

Table 1
Data statistical analysis.

Variable	With disease (n=507)	Without disease (n=410)	P-value
Age	55.876 ± 8.7	50.551 ± 9.4	<0.001
Sex (male)	457 (90 %)	267 (65 %)	<0.001
chest pain type (asymptomatic)	392 (77 %)	149 (25 %)	<0.001
resting bp s	134.154 ± 19.8	130.180 ± 16.5	0.001
Cholesterol	250.091 ± 52.9	238.425 ± 54.1	0.001
fasting blood sugar (false)	338 (68 %)	366 (89 %)	<0.001
resting ecg (normal)	284 (56 %)	267 (65 %)	0.004
max heart rate	127.647 ± 23.4	148.151 ± 23.3	<0.001
exercise angina (yes)	316 (62 %)	55 (13 %)	<0.001
oldpeak	1.277 ± 1.2	0.408 ± 0.7	<0.001
ST slope (flat)	380 (75 %)	79 (19 %)	<0.001

presented in our dataset is diversified. Diversity plays a critical role when the problem involves ML. It helps to ensure the model's performance and will promote its capacity to generalize when the classifier is executed in a real-life context. Gong et al. [24] show that diversity is essential to improve the representational ability of the model and Mehrabi et al. [25] presented examples of the potential real-world problems that non-diverse data can cause, and why fairness is an essential factor.

To compare the two groups (patients with and without disease) and see if the values are related, we have used the T-test, also known as Student's test, in numeric variables [26]. For nominal and binary features, chi square ($\chi 2$) test was applied [27]. Table 1 shows that all features have a P-value < 0.05, hence rejecting the null hypothesis [28]. In the table, each numeric feature presents the mean value $\pm$ standard deviation and non-numeric data is expressed as number and percentage. The same approach was used in Wu et al. work [29].

2.2. Experimental approach

2.2.1. Data preprocessing

Data is an essential resource to promote the performance of ML models and it can be in different formats. Data preprocessing is a fundamental practice that transforms data into valuable information that will then be interpreted by an ML model. Alexandropoulos et al. [30] shows the benefits of data preprocessing in well-known ML classifiers and how the application (or not) of the preprocessing techniques can directly affect their performance. In this work, the authors applied different data cleaning and data transformation techniques. Domain knowledge is essential when dealing with clinical data. Therefore, all decisions made here were based on the data itself and what is recommended in the literature. Proper data modeling is equally important, as it ensures that the data structure and relationships are accurately represented, which enhances the model's ability to learn and make accurate predictions [16]. Several works present similar preprocessing steps to the ones applied in this work [31–34].

2.2.1.1. Invalid data. We detected that one instance of the feature ST Slope, which is the code given for the slope of the peak exercise ST segment, was invalid. Based on the dataset documentation this is a nominal feature that varies between 1 and 3. An instance with a zero value was found. Therefore, because it was just one instance, we decided to delete it from the dataset and work with the remaining 1189 instances. After that, the authors noticed that 172 instances had a cholesterol value of zero, which is clinically impossible because human beings must always have a positive value of cholesterol. To correct this, we trained a Random Forest (RF) regression model on the known data and used this model to predict and impute cholesterol values on the invalid instances as indicated by Tang and Khan et al. [35,36].

2.2.1.2. Duplicate data. This dataset results from merging five other datasets. The authors noticed that there were 272 repeated instances and proceeded to remove them. Duplicate entries can ruin the split between train, validation, and test sets. To avoid that, this process was done on the entire dataset. This is a critical point to solve in every ML problem since it can lead to overfitting.

2.2.1.3. Normalization. Normalization is a technique used to rescale numeric columns to be on the same scale without losing information. We decided to apply normalization with the Z-score method. This method was recommended by Singh et al. [37] who presented a wide comparison between fourteen different normalization methods and their respective effect on the classifier performance. The results show that normalization brings improvements to the classification process and the study also proves the effectiveness of using the Z-score method.

2.2.1.4. Feature transformation. In the data analysis stage, we often find that some numerical data features can be nonGaussian and asymmetric, which complicates data analysis. Feature transformation changes the shape of the data distribution while keeping its information. The method used in this process was the yeo-johnson that changes the feature set to make it follow a Gaussian-like or normal distribution [38]. This method was suggested by Batista and Toupas et al. [39,40]. Feature selection techniques were not used because the low data dimensionality does not imply reducing the number of input features [41], and the classifiers used are tree-based models. This kind of model is capable of selecting the optimal features to use by measuring the information gain criterion, which can be defined as the amount of information that is obtained by selecting certain features in favor of others [42].

2.2.1.5. Handling outliers. Outliers from the training data were detected and removed through Principal Component Analysis (PCA) linear dimensionality reduction using the Singular Value Decomposition technique. A threshold of 0.05 was chosen, this means that 0.025 of the values on each side of the distribution's tail are removed from the training set as indicated by [43].

2.2.1.6. Collinear features. Collinear features are variables that have a high correlation with each other. This means that an independent variable can be predicted from another independent variable. Having collinear features can increase the variance of the coefficients, making linear models unstable and noisy [44]. If collinear variables are detected, one of the two must be dropped in preprocessing. The multicollinearity threshold used was 0.9, which means almost perfect multicollinearity between independent variables. No collinear variables were detected.

2.2.1.7. Data imbalance. After all the previous preprocessing steps, there were 840 instances left to train the model, having an imbalance ratio of 0.79. Although this corresponds to a relatively well-balanced distribution of the class variable, the authors decided

to perfectly balance the data using Synthetic Minority Oversampling Technique (SMOTE) and analyze the gain in the performance of the classifiers. The results show a small improvement in almost all the classifiers' performance when SMOTE was applied. Furthermore, SMOTE is one of the main methods to generate data, being widely addressed and recommended in the literature [45].

2.2.2. ML configuration

This study was developed using PyCaret in version 2.3.5, which is an open-source and low-code ML library in Python that is used to automate ML workflows. PyCaret is a wrapper with several different ML libraries and frameworks such as scikit-learn, XGBoost, CatBoost, spaCy, LightGBM, Optuna, Hyperopt, and Ray [46]. Python version 3.7.13 was used to make all the necessary code. After the preprocessing stage was done, we started by testing fifteen different ML classifiers: CatBoost, RF, Extra Trees (ET), Ridge, Linear Discriminant Analysis, Extreme Gradient Boosting, Logistic Regression, Light Gradient Boosting Machine, Gradient Boosting, Naive Bayes, k nearest neighbor (KNN), AdaBoost, Decision Tree, Support Vector Machine (SVM) and Quadratic Discriminant Analysis. All these classifiers were tested with their default hyper-parameters and the results showed that the three best classifiers in terms of Sensitivity were RF, ET, and CatBoost.

The three best models are tree-based models, that is, a type of ML algorithm that partitions the training data into subsets using sequences of conditional statements. Each successive split increases the model's complexity, which can then be used in prediction [47]. RF is an ensemble of decision trees, where a random subset of the available features is used to choose the optimal feature to make the split at each node. Denisko et al. [48] explain the behavior of this classifier in detail. While RF split nodes based on the optimum split, ET chooses it randomly. If we look at the ET algorithm, all the training set is used to train each regression tree. On the other side, a bootstrap replica is used to train the RF model [49]. In turn, CatBoost is a gradient boosting algorithm, it is capable of processing categorical features and implements ordered boosting, which is a permutation-driven alternative to the approach used in the classic boosting algorithm [50].

The performance of ML classifiers depends on their hyperparameter settings. This process is called hyperparameter or model tuning and its critical impact is demonstrated by Probst et al. [51]. In our study, three different python libraries were used for tuning: Scikit-Learn, Optuna, and scikit-optimize. The Scikit-Learn library was introduced by Pedregosa et al. in 2011 [52] as a Python package that incorporates numerous cutting-edge ML techniques for both supervised and unsupervised problems. It was used with the default RandomizedSearchCV algorithm. This randomly passes the set of hyperparameters and uses a score to give the best set of hyperparameters [53]. OptunaSearchCV was implemented through the Optuna Library [54], it works similarly to the RandomizedSearchCV but it applies cross-validation by default with $k = 5$. Finally, from scikit-optimize [55], we used BayesSearchCV, which is a Bayesian optimization. It consists of a method based on the Bayes theorem to search for a global optimization problem efficiently and effectively [56]. All tuning algorithms were tested, and we chose the best for each classifier. Catboost and RF had better performance with the RandomizedSearchCV algorithm and ET had better results with the OptunaSearchCV.

Table 2 shows the hyperparameters resulting from the application of these optimization techniques. The default parameters for each of these models can be seen in Appendix A.

2.2.3. Experimental setup for evaluation

All the experimental setup was made using the PyCaret framework which uses several built-in ML libraries and frameworks like scikit learn, XGBoost, CatBoost, and others. This tool is one of the best and most useful low-code autoML frameworks as suggested by Gain et al. [57]. Each model performance was measured using the following metrics: accuracy, Sensitivity, Area under the ROC Curve - AUC, Precision, and F1 Score.

Initially, the data was split into two distinct sets, 5 % for validation/unseen data and 95 % for training and testing. Data in the train and test dataset were split into a 3:7 ratio, 30 % for testing and 70 % for training. This is the recommended ratio for splitting data into training and testing as statistically proven by Gholamy et al. [58]. Of the 840 instances we have, 262 are used for testing and 578 for training.

From the fifteen classifiers initially tested, we selected the three best ones based on their performance on the training set. To

Table 2
Hyperparameters used in the proposed models.

Model	Hyperparameters	Values
Catboost	Learning rate	0.0010
	Depth	10
	Max leaves	1024
	Iterations	250
	Evaluation metric	Logloss
	Score function	Cosine
Random Forest	Criterion	Entropy
	Max depth	4
	Estimators	70
	Max features	Sqrt
Extra Trees	Criterion	Entropy
	Max depth	4
	Estimators	289
	Max features	0.4097

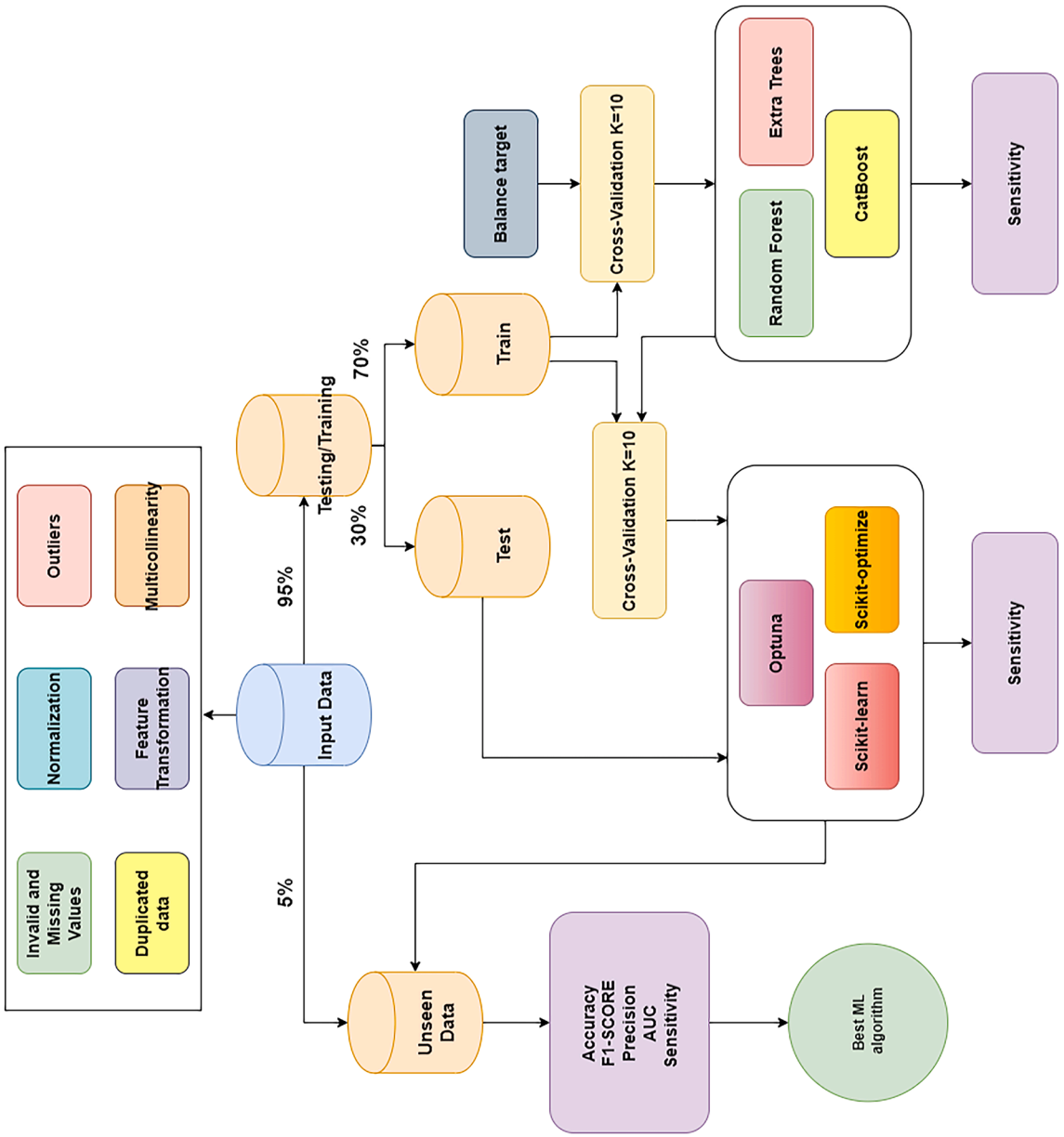


Fig. 1. Experimental setup.

evaluate performance, the Sensitivity metric was used, this being the most suitable and recommended when we are working in the medical domain [59–61]. To train the models we used cross-validation with K = 10, which means that for every K-fold, 9/10 of the data is used for training and 1/10 is used for testing. Although we used cross-validation to train the models, it was necessary to split the data into training and testing because cross-validation should only be applied to the 578 training instances to ensure that the model is tested on data never seen before in the training process. This ensures that there is no occurrence of data leakage, in other words, data previously used in training that is also used in the test, which can lead to a false impression of the generalizability of the model [62]. Using k-fold cross-validation, the oversample of the data with SMOTE occurs on the training data for each iteration. At each K-fold, the minority class is augmented creating a perfectly balanced data (imbalance ratio = 1). This prevents overoptimistic results and is the recommended method to validate models on this type of problem [63].

The three best models selected based on performance on the training set were then subjected to the parameter tuning process. The authors used and tested different tuning libraries and their algorithms to achieve the best possible result for each classifier. Scikit-Learn library was used with the RandomizedSearchCV algorithm, Optuna library with OptunaSearchCv, and Scikit-optimize library with the BayesSearchCV algorithm. Optimal parameters were chosen using a training set and cross-validation with K = 10. It is a common practice to use cross-validation in the parameter tuning phase [64]. The authors have, once again, focused on sensitivity to choose the best hyperparameter optimization for each model. The best combinations of classifiers and tuning methods were chosen based on the performance in training and their capacity to generalize in the test. Models that provide a satisfactory performance on the training set but then have a considerable drop on the test set reveal a sign of possible overfitting [65]. After obtaining the three different models with the optimized parameters, they were trained on the total data set by joining test and training data (840 instances).

The selection of the best model among these three already optimized classifiers is done according to the performance that the various models obtained in the 5 % of the data set which was never used before in the training and testing process. The unseen data set was almost perfectly balanced in the class to ensure the veracity of the results. All the steps done in the experimental setup of this study are represented in Fig. 1.

3. Results

3.1. Performance of predictive models

In the healthcare domain, the most significant aspect of an ML model is the ability to detect the people who are sick. If we define a positive example as a “person who has a disease”, the detection of an actual negative as a positive is not the worst case. However, if we detect an actual positive patient as a negative, we are ignoring a positive case and exposing the person’s health since this patient can leave the monitoring process. Consequently, the most significant metric for our analysis is Sensitivity. It is a performance metric widely used in medical studies [59–61]. Initial fifteen ML models were trained using cross-validation with $k = 10$ and the results are presented in

Table 3. RF, Catboost, and Extra Trees obtained the best result and were selected.

The selected models tuned with the hyperparameter optimization algorithms were evaluated. The respective sensitivity values are presented in Table 4 for the train set and Table 5 for the test/hold-out set. The results showed that the best models on the train set tend to show signals of overfitting on the test set due to the considerable decrease in the sensitivity value. Models that have remained constant or even increased their performance between training and testing are more reliable. Catboost and ET had the best hyperparameter optimization with the Optuna - OptunaSearchCV. RF classifier has the best result with Scikit-Optimize - BayesSearchCV.

The three best-optimized models were selected: BayesSearchCV optimized RF, OptunaSearchCV optimized ET, and OptunaSearchCV optimized Catboost. The results obtained in Table 6 show the reliable performance of the three classifiers on unseen data, and we can state that RF was the model that has the best performance with 95.65 % of accuracy and 100 % sensitivity.

The receiver operating characteristic (ROC) analysis is a well-known measure for evaluating classifiers’ performance [66]. It is

Table 3
Initial results on train set with cross-validation.

Model	Accuracy	AUC	Precision	Sensitivity	F1 Score
Random Forest	0.8719	0.9247	0.8821	0.9049	0.8820
Catboost	0.8789	0.9350	0.8939	0.9021	0.8970
Extra Trees	0.8685	0.9261	0.8812	0.8993	0.8891
Light gradient boosting	0.8564	0.9168	0.8992	0.8654	0.8802
Gradient boosting	0.8616	0.9213	0.8933	0.8747	0.8830
Extreme gradient boosting	0.8356	0.9139	0.8755	0.8523	0.8620
Naive Bayes	0.8529	0.9125	0.8727	0.8765	0.8743
Ridge	0.8581	0.0000	0.8607	0.8903	0.8762
Linear Discriminant analysis	0.8581	0.9336	0.8607	0.8833	0.8762
K nearest neighbors	0.8547	0.9029	0.8579	0.8809	0.8731
Logistic Regression	0.8581	0.9304	0.8578	0.8866	0.8763
Decision Tree	0.8321	0.8280	0.8519	0.8618	0.8553
Ada Boost	0.8321	0.8983	0.8430	0.8702	0.8547
SVM – Linear kernel	0.8304	0.0000	0.7810	0.8860	0.8418
Quadratic discriminant analysis	0.4152	0.5000	0.0000	0.0000	0.0000

Table 4
Tuned models sensitivity on the train set.

Model Tuning method	Catboost	Random Forest	Extra Trees
Scikit-Learn – RandomizedSearchCV	**0.9052**	**0.9111**	0.8843
Optuna – OptunaSearchCV	0.8932	0.8873	0.8933
scikit-optimize – BayesSearchCV	0.8754	0.9082	**0.8964**

Table 5
Tuned models sensitivity on test/hold-out set.

Model Tuning method	Catboost	Random Forest	Extra Trees
Scikit-Learn – RandomizedSearchCV	0.8837	0.8837	0.8682
Optuna – OptunaSearchCV	**0.8930**	0.8837	**0.8834**
scikit-optimize – BayesSearchCV	0.8915	**0.9170**	0.8682

Table 6
Tuned models' performance on unseen data.

Model	Accuracy	AUC	Precision	Sensitivity	F1 Score
Tuned Random Forest	0.9565	0.9754	0.9231	1	0.9600
Tuned Catboost	0.9348	0.9792	0.9200	0.9593	0.9388
Tuned Extra Trees	0.8696	0.9735	0.8462	0.9167	0.8800

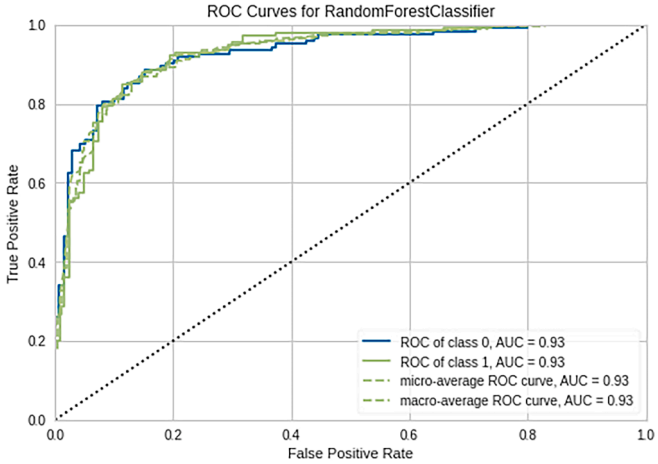
based on specificity (also called False Positive Rate - FPR) and sensitivity (also called True Positive Rate - TPR) performance metrics. To plot the ROC curve, it is considered TPR (y-axis) against FPR (x-axis). The curve shows the trade-off between sensitivity and specificity. This trade-off is represented in Fig. 2(a-c) for Catboost, RF, and ET, respectively. If we have TPR equal to FPR it means that the model produces the same number of false and true positives, which makes the classifier ineffective. The closer to the upper left corner our curve is, the better our model performs [67]. The ROC curves of the proposed classifiers are very similar, it is visible that the RF curve is slightly closer to the upper left corner.

3.2. Explanation of the machine learning model

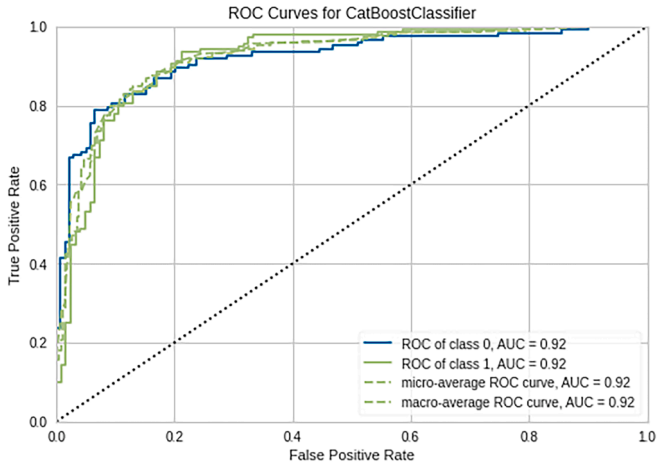
Most of the best performance ML models are Black Boxes, which means that it is difficult to explain to the readers the results obtained from the model prediction. It is critical to have interpretable models to provide data-driven and well-informed decisions in healthcare. Stiglic et al. propose an overview of several different interpretability approaches and examples where they can be applied in the healthcare areas [68]. Over the years there has been a trade-off between interpretation and accuracy since simple models are easy to interpret but in return, the accuracy tends to decrease. Instead, if we increase complexity, accuracy values will also increase but it is extremely difficult to interpret the results obtained. Consequently, we used SHapley Additive exPlanations, known as the SHAP method. This method was proposed by Lundberg et al. as a unified method for analyzing model predictions [69]. In short, the SHAP method explains the importance of each feature to an individual's predicted value. In our case, one feature can either have a positive or a negative influence on a given patient to be classified as healthy or sick. With this method, the authors provide clear information on the feature contributions for each observation (local) or a set of observations (global). Fig. 3 shows the pipeline used from the input model to have an explanation of the respective output.

The authors have used Pycaret built-in Shapley Additive exPlanations - SHAP functions that analyze the predictions generated from the trained model to explain the results of our best model (Tuned RF). Fig. 4 presents a global explanation represented by a Beeswarm graph. All variables are shown on the Y-axis in the order of global feature importance, the first one is the most significant and the last is the least significant one. Before preprocessing, the dataset contained twelve features but after data transformation, the number of features increased to eighteen. Looking at the X-axis of the graph, we can see those values further to the left of the graph have a more negative impact on prediction while values further to the right have a more positive contribution. Red dots indicate high values of the respective feature, while blue dots indicate a low value. The most significant variable presented is ST slope_1. Furthermore, it is possible to notice that High values of ST slope_1 lead to a negative contribution in the prediction while low values lead to a positive contribution. On the other hand, looking at the chest pain type_4 variable, we can say that high values contribute to a positive prediction and low values have a negative impact. Chest pain type 1, 2, and 3, ST slope 3, and both resting ECG 1 and 2 had minimal impact on the model predictions.

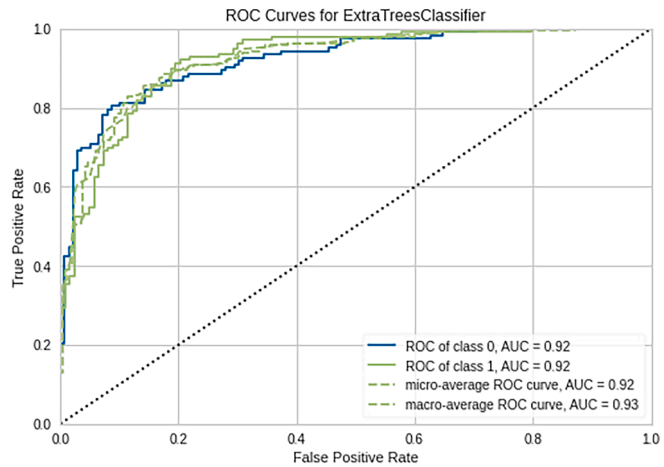
Although global explanations are critical, local explanations make it possible to explain a given prediction made by the model. Fig. 5 shows a randomly chosen prediction made by the proposed model on test data. If f(x) calculated is above 0.5 it is a positive prediction, so in this case, f(x) is equal to 0.789 which shows us that this is a patient with heart disease. To understand the model prediction, it is necessary to interpret the local explanation and relate it to the global explanation shown above. Chest pain type 4, cholesterol, max heart rate, and oldpeak were the variables that contributed most to the positive prediction of the model. The global



(a)



(b)



(c)

Fig. 2. ROC Curves.

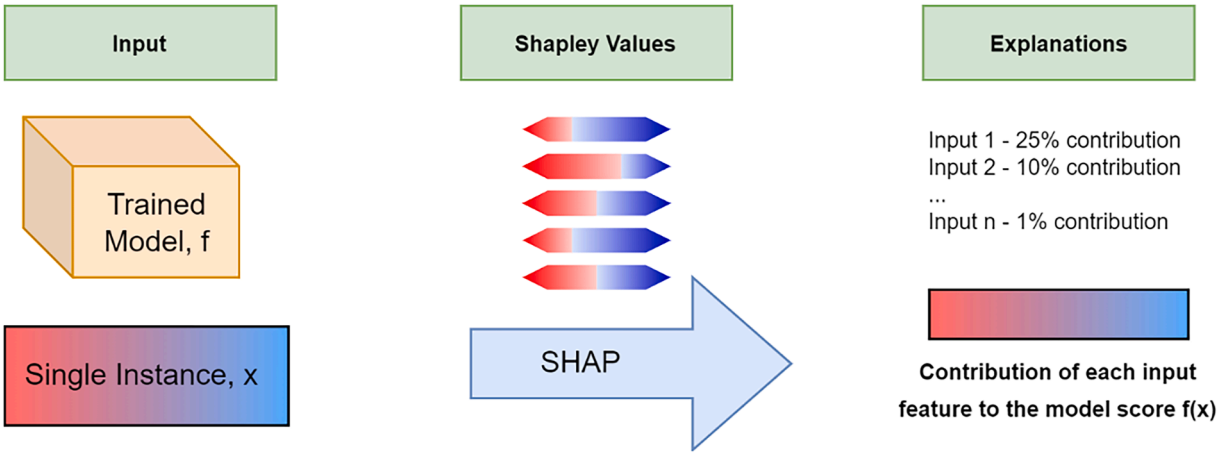


Fig. 3. Pipeline used for interpretability.

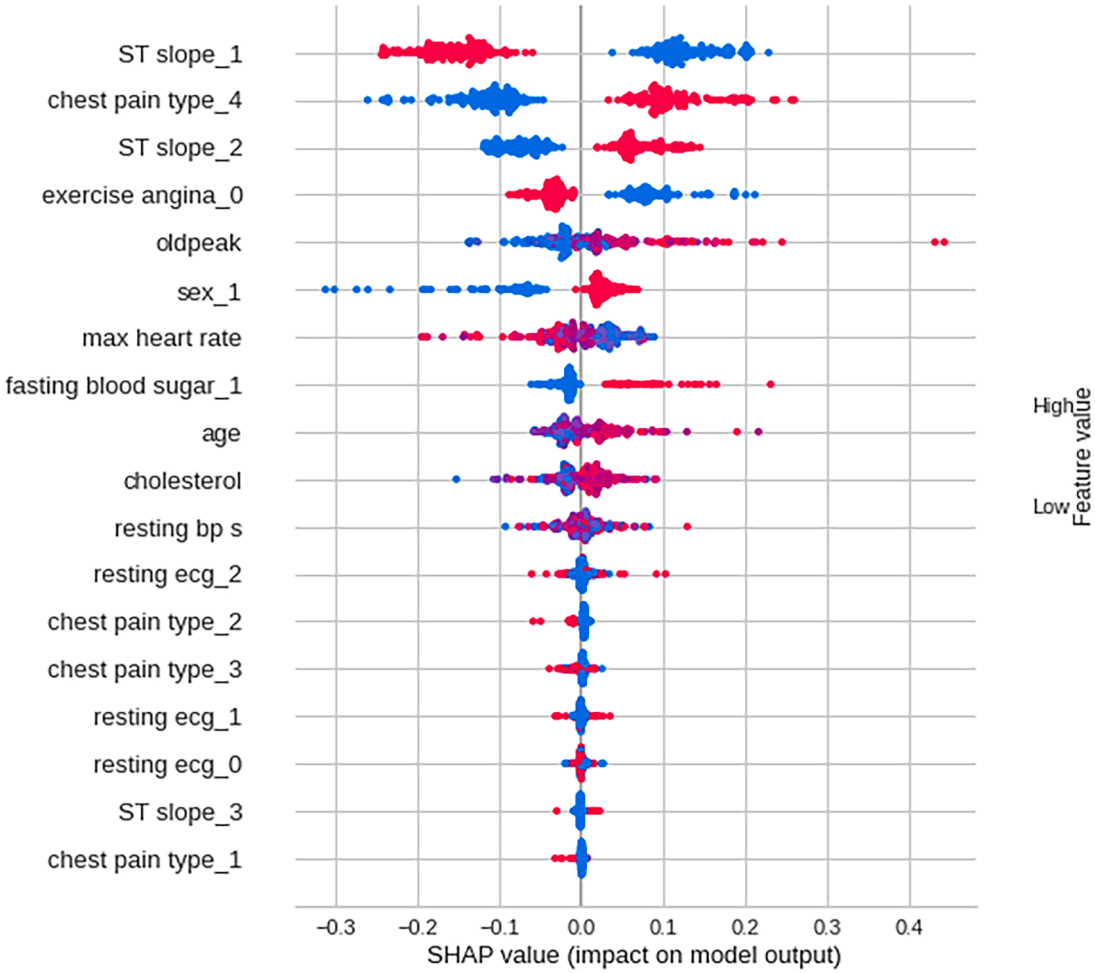


Fig. 4. Beeswarm graph - global explanation.

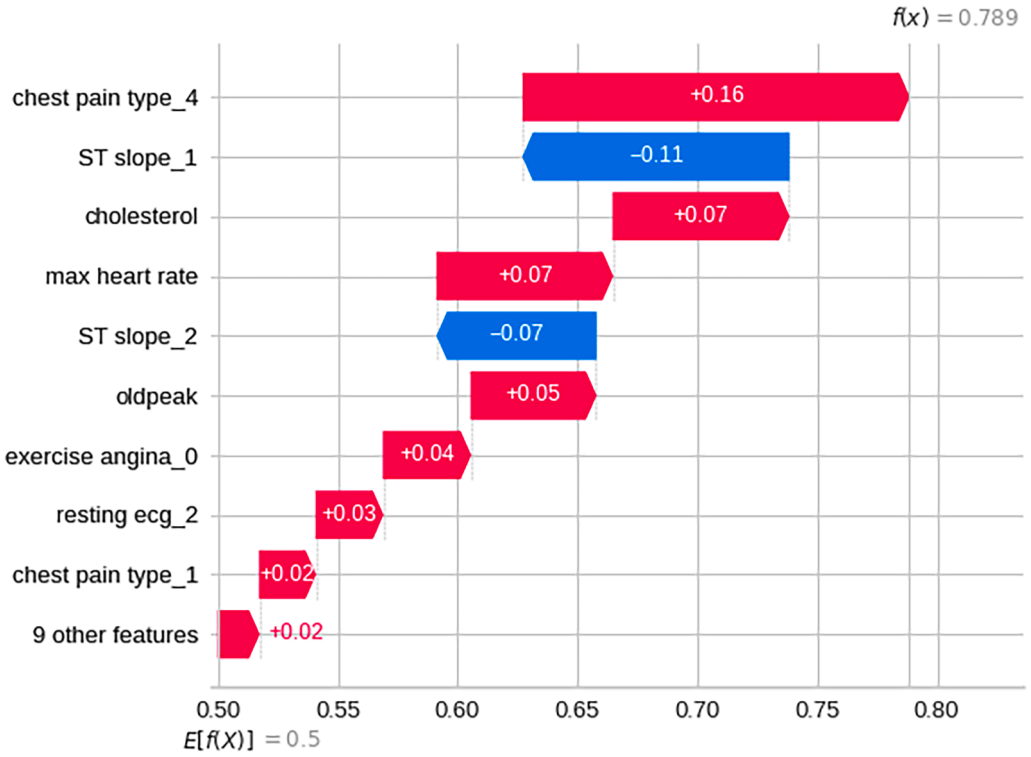


Fig. 5. Waterfall graph - local explanation.

explanation graph shows with some certainty that this patient has Chest pain of type 4, which is an asymptomatic type of pain, a low value of max heart rate, and a high value of oldpeak and cholesterol. Contrastingly, the patient has an ST slope of type 1 which means that the slope of the peak exercise ST segment is rising and consequently has no ST slope of type 2, both variables had a negative impact on the prediction, but it was not enough for the model to predict that the patient was healthy.

4. Discussion

In this study, several ML algorithms were applied to the problem of heart disease aiming to automate the process of detecting positive cases, enabling faster detection of early-stage patients, and increasing the chances of survival. In medicine, any decision made must always be based on information that can be interpreted by doctors. This is particularly critical to support the doctor's decision and explanation to the patient. Therefore, it is mandatory to not only predict correctly but also be interpreted and explained.

The authors started to work by applying several data preprocessing techniques, and then three models with the best initial results were selected. Several tuning techniques have been used to improve their results, and finally, the best model has been selected. Furthermore, the authors prove that it is possible to interpret the predictions using SHAP. The best model was RF tuned with the Bayesian optimization technique provided by the Scikit Optimize library. Firdaus et al. [70] show that different tuning algorithms applied to a similar heart disease dataset and Bayesian optimization had not only the best result but also the shortest tuning time. Model train and test results were compared to ensure that there was no overfit and after that, it was also validated on unseen data. The results of the proposed model obtained on the unseen data set were 95.65 % accuracy and 100 % sensitivity. The dataset used in this work consists of a combination of five different heart datasets. When comparing the proposed method with other studies on the same dataset, we can see that the results are promising.

Yilmaz et al. [7] used an SVM variation of SMOTE to balance this dataset. Then the data was split, 80 % and 20 % for testing and training, respectively. Logistic Regression, SVM, and RF classifiers were used and their respective hyperparameter values were determined by grid search. The proposed model proposed was RF, it obtained 92.9 % of accuracy and 92.8 % of sensitivity, 2.75 % less accuracy, and 7.2 % less sensitivity when compared to our results. This study also does not provide information about pre-processing done on the data such as the removal of duplicates, removal of outliers, or feature transformations.

To make a model capable of identifying cardiovascular diseases, Alalawi et al. [71] used a CVD dataset and the heart disease dataset. The authors tried an approach based on feature selection but found that the results do not improve. This approach differs greatly from our approach since no data balancing techniques were used, and there was no information about how the hyper-parameters for each model were chosen or how exactly the data were preprocessed. The results obtained were also low, where the best model was gradient boosting with 70 % accuracy and 70 % sensitivity.

Doppala et al. [72] proposed an ensemble model to also identify CVD. The dataset was balanced, pre-processed and a voting

ensemble method was proposed. This method consists of the combination of multiple other models, in this case, the models aggregated were Naive Bayes, RF, SVM, and Gradient Boosting. It uses a voting mechanism to decide the output. It achieved 93.39 % accuracy and 94 % sensitivity, less 2.26 % accuracy, and 6 % sensitivity when compared to our proposed method.

To our best knowledge, no similar study that used the same dataset has tried to explain the predictions made by the model. However, other relevant works are using heart disease datasets. Wang et al. [73] provide an extensive approach to the interpretability of an ML model to predict mortality in patients caused by coronary heart disease, more specifically, heart failure. SHAP was used to make some analysis and interpret the results. The gender-based analysis led to interesting conclusions. Although this work presented a complete model interpretation, we cannot make a direct comparison in terms of performance as the dataset differs greatly from the one used in our study.

An approach using intelligent optimization algorithms was made by Khourdifi et al. [74]. The Fast Correlation-Based Feature Selection (FCBF) method was used to filter redundant features and Particle Swarm Optimization (PSO) combined with Ant Colony Optimization (ACO) was used for feature optimization. The best result obtained with these optimizers was achieved with KNN, reporting 100 % sensitivity and 99.65 % accuracy. Khourdifi et al. achieved the same sensitivity and better accuracy than our proposed method. However, our study presents an explanation for the model output, making it as close as possible to a white-box model.

Table 7 shows the complete comparison between the proposed method and studies found in the literature. This comparison is possible because most of the compared papers use the same dataset as us or its variation. The papers that differ in the dataset still present similar information and predict the same outcome, that is, whether or not the person has heart disease. Other works use two different variations of our dataset, both coming from the Hearth Disease Database available in the UCI Machine Learning Repository which contains 303 records and 75 features [75]. Miao et al. [76] used this data but they only selected 29 features due to the number of missing values. On the other hand, Khourdifi et al. [74] and Anitha et al. [77] used a subset of the dataset, the Cleveland Clinic Foundation dataset that contains 270 instances and 13 features.

Studies show that there is usually a trade-off between interpretability and performance [78,79]. A simple model is easily interpretable but has no extraordinary results while a complex model despite having superior performance becomes almost impossible to explain. We address this problem by using SHAP. The proposed model outperforms most of those presented in the literature, and at the same time, we were able to interpret how the model gives a particular prediction using the SHAP values. Models with high performance and interpretability are essential for sensitive areas such as the medical field [17].

The main contributions of this study are described as follows. First, we proposed an automated and efficient way to predict heart disease using ML algorithms with reliable performance. Contrary to what is usually found in the literature, all details necessary to replicate the work were shared ensuring that the model is as transparent as possible [80]. The model was validated using the three-Way Holdout Method for Hyperparameter Tuning, similar to what is presented by Raschka [81]. Second, three different tuning libraries and algorithms were tested and compared. The best combination was chosen for each model and details about the hyperparameters were presented. Finally, the interpretability of ML models has been widely discussed.

5. Threats to validity

The validity of an experimental study can be divided into two large groups: internal validity and external validity. The first aims to understand if there is enough evidence to support our claims. In the case of external validity, it focuses on the generalizability that the study may have when applied to another sample [82].

Internally, we tried throughout our study to minimize all the threats we detected, always presenting the criteria behind each decision. Many of the papers presented in the medical field literature lack the details necessary to ensure their replicability [83], which in itself is a major threat to their validity. In our work, all actions taken are clearly described and replicable. Threats like data leakage, sampling bias, a non-random split of data, overfitting, and data imbalance were identified. After that, several techniques were applied to reduce their impact as much as possible. As with any study, there are of course still potential risks for internal validation. Future studies should be done to make sure that there is no bias in the proposed algorithms to ensure that the results presented are fair, and free of any kind of discrimination [84]. Domain knowledge limitations can also be a potential risk, especially when making decisions during data preprocessing and ML model building. However, the authors have used existing literature to support all decisions made.

To lower the risks related to external validation, we use different methods, including holdout, cross-validation, and a final validation set containing data never seen in training. In addition, the interpretation of the model was provided, making it possible for any physician to understand and validate the prediction made. Still, the inability to generalize continues to affect ML predictive models. There are several threats to the generalizability of a predictive model in healthcare that may lower its performance, such as: (i) Patient

Table 7
Results comparison.

Authors	Model	Accuracy	AUC	Precision	Sensitivity	F1	Interpretable
Yilmaz et al. [7]	RF	0.929	-	0.929	0.928	0.928	No
Alalawi et al. [71]	GB	0.700	-	0.700	0.700	0.700	No
Doppala et al. [72]	Ensemble	0.934	-	0.990	0.940	0.900	No
Khourdifi et al. [74]	Optimized KNN	0.996	-	0.993	1	0.997	No
Anitha et al. [77]	Naive Bayes	0.867	-	0.857	0.811	0.834	No
Miao et al. [76]	DNN	0.836	0.892	0.791	0.935	0.857	No
Proposed Method	**Tuned RF**	**0.956**	**0.975**	**0.923**	**1**	**0.9600**	**Yes**

demographic variation (ii) Patient genotypic and phenotypic variation (iii) System and methodology used to obtain the data (iv) Changes in practice pattern over time and (v) Variation in secondary factors such as political, environmental, and cultural [85]. This is, therefore, a major challenge for predictive models, where the demand for generalizability leads most often to models with low performance, creating a trade-off between high-performance models for a specific group or low-performance models but with some generalization capacity.

The development of high-performance and explainable heart disease classifiers is a major challenge in ML. Therefore, this study has some limitations. In the preprocessing phase, other procedures and methods could be explored and tested. Different balancing techniques could be used, for example, different variations of SMOTE [86] or Balanced Class Weight [87]. Another limitation is that the selection of the ML models was not made by any specific criteria such as the size of the data or the representativeness of the sample itself. Besides that, other classifiers could have been used, such as neural networks. Furthermore, a complete comparison with other models in the literature becomes challenging because most of the papers do not provide essential information such as the pre-processing steps or the values of the hyperparameters. Finally, the model has been tested on a comprehensive dataset, however, other datasets are available and like any other ML model, it is not possible to guarantee that the performance will be maintained when applied in a real context. In future work, the authors intend to make the model available online using MLaaS (Machine Learning as a Service) based architectures. To this end, we intend to explore solutions such as MLCapsule [88] which is a guarded offline deployment of MLaaS. This will help us to be able to test the model in a real context to understand its real value.

6. Conclusions

CHD is the major cause of mortality worldwide and is likely to get even worse in the coming years. Despite numerous proposed ML models for predicting heart disease, those that perform best are often difficult to interpret, leading to a lack of user confidence and making their practical implementation unfeasible. This study shows that is not only possible to have a model with reliable performance but also to interpret their predictions by using SHAP values. Data preprocessing and modeling were detailed, and various tuning techniques were explored to optimize the parameters for each classifier, ensuring the development of the best possible model. The OptunaSearchCV algorithm had the best tuning results on Catboost and ET, but the proposed classifier, RF, achieved the best results using a Bayesian optimization technique - BayesSearchCV. The findings indicate not only the feasibility of developing a predictor with consistent performance but also the potential for comprehensive interpretation of prediction results. All the details of what was done throughout the work have been provided. The data used is public, and all the data preprocessing methods are shown and explained. Additionally, the complete workflow involved in constructing predictive models, including the fine-tuned hyperparameters, is widely explained. Consequently, the work is easily replicable, offering valuable support for future research seeking proficient and interpretable predictive models.

CRediT authorship contribution statement

Francisco Mesquita: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization. **Gonçalo Marques:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.datak.2024.102339.

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